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import java.sql.*;
import java.util.Scanner;

public class BankTransactionManager {
    private static final String URL = "jdbc:mysql://localhost:3306/bank_db";
    private static final String USER = "root";
    private static final String PASSWORD = "password";

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        try (Connection conn = DriverManager.getConnection(URL, USER, PASSWORD))
        {
            while (true) {
                System.out.println("\n--- Bank Account Transaction Management
                ---");

                System.out.println("1. Create Account");
                System.out.println("2. Deposit Money");
                System.out.println("3. Withdraw Money");
                System.out.println("4. Display Account Details");
                System.out.println("5. Exit");
                System.out.print("Enter choice: ");

                String choice = scanner.nextLine().trim();
                switch (choice) {
                    case "1":
                        createAccount(conn, scanner);
                        break;
                    case "2":
                        deposit(conn, scanner);
                        break;
                    case "3":
                        withdraw(conn, scanner);
                        break;
                    case "4":
                        displayDetails(conn, scanner);
                        break;
                    case "5":
                        System.out.println("Exiting application.");
                        return;
                    default:
                        System.out.println("Invalid choice. Try again.");
                }
            }
        } catch (SQLException e) {
            System.out.println("Database connection error: " + e.getMessage());
        }
    }

    private static void createAccount(Connection conn, Scanner scanner) {
        try {
            System.out.print("Enter Account Number: ");
            int accNum = Integer.parseInt(scanner.nextLine().trim());
            System.out.print("Enter Account Holder Name: ");
            String name = scanner.nextLine().trim();
            System.out.print("Enter Initial Balance: ");
            double balance = Double.parseDouble(scanner.nextLine().trim());

            String sql = "INSERT INTO accounts (account_number, holder_name,
            balance) VALUES (?, ?, ?)";
            try (PreparedStatement pstmt = conn.prepareStatement(sql)) {
                pstmt.setInt(1, accNum);
                pstmt.setString(2, name);
                pstmt.setDouble(3, balance);
                pstmt.executeUpdate();
            }
        }
    }
}

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        System.out.println("Account created successfully.");
    }
} catch (NumberFormatException e) {
    System.out.println("Error: Invalid number format input.");
} catch (SQLException e) {
    System.out.println("Database error: " + e.getMessage());
}
}

private static void deposit(Connection conn, Scanner scanner) {
    try {
        System.out.print("Enter Account Number: ");
        int accNum = Integer.parseInt(scanner.nextLine().trim());
        System.out.print("Enter Deposit Amount: ");
        double amount = Double.parseDouble(scanner.nextLine().trim());

        if (amount <= 0) {
            System.out.println("Deposit amount must be positive.");
            return;
        }

        String selectSql = "SELECT balance FROM accounts WHERE
account_number = ?";
        String updateSql = "UPDATE accounts SET balance = balance + ? WHERE
account_number = ?";

        try (PreparedStatement selectPstmt =
conn.prepareStatement(selectSql);
            PreparedStatement updatePstmt =
conn.prepareStatement(updateSql)) {

            selectPstmt.setInt(1, accNum);
            try (ResultSet rs = selectPstmt.executeQuery()) {
                if (rs.next()) {
                    updatePstmt.setDouble(1, amount);
                    updatePstmt.setInt(2, accNum);
                    updatePstmt.executeUpdate();

                    double newBalance = getBalance(conn, accNum);
                    System.out.println("Balance = " + newBalance);
                } else {
                    System.out.println("Account not found.");
                }
            }
        }
    } catch (NumberFormatException e) {
        System.out.println("Error: Invalid input.");
    } catch (SQLException e) {
        System.out.println("Database error: " + e.getMessage());
    }
}

private static void withdraw(Connection conn, Scanner scanner) {
    try {
        System.out.print("Enter Account Number: ");
        int accNum = Integer.parseInt(scanner.nextLine().trim());
        System.out.print("Enter Withdrawal Amount: ");
        double amount = Double.parseDouble(scanner.nextLine().trim());

        if (amount <= 0) {
            System.out.println("Withdrawal amount must be positive.");
            return;
        }
    }
}

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        String selectSql = "SELECT balance FROM accounts WHERE
account_number = ?";
        String updateSql = "UPDATE accounts SET balance = balance - ? WHERE
account_number = ?";

        try (PreparedStatement selectPstmt =
conn.prepareStatement(selectSql);
            PreparedStatement updatePstmt =
conn.prepareStatement(updateSql)) {

            selectPstmt.setInt(1, accNum);
            try (ResultSet rs = selectPstmt.executeQuery()) {
                if (rs.next()) {
                    double currentBalance = rs.getDouble("balance");
                    if (amount > currentBalance) {
                        System.out.println("Insufficient Balance");
                    } else {
                        updatePstmt.setDouble(1, amount);
                        updatePstmt.setInt(2, accNum);
                        updatePstmt.executeUpdate();

                        double newBalance = getBalance(conn, accNum);
                        System.out.println("Balance = " + newBalance);
                    }
                } else {
                    System.out.println("Account not found.");
                }
            }
        } catch (NumberFormatException e) {
            System.out.println("Error: Invalid input.");
        } catch (SQLException e) {
            System.out.println("Database error: " + e.getMessage());
        }
    }

    private static void displayDetails(Connection conn, Scanner scanner) {
        try {
            System.out.print("Enter Account Number: ");
            int accNum = Integer.parseInt(scanner.nextLine().trim());

            String sql = "SELECT * FROM accounts WHERE account_number = ?";
            try (PreparedStatement pstmt = conn.prepareStatement(sql)) {
                pstmt.setInt(1, accNum);
                try (ResultSet rs = pstmt.executeQuery()) {
                    if (rs.next()) {
                        System.out.println("Account Number: " +
rs.getInt("account_number"));
                        System.out.println("Account Holder: " +
rs.getString("holder_name"));
                        System.out.println("Balance: " +
rs.getDouble("balance"));
                    } else {
                        System.out.println("Account not found.");
                    }
                }
            }
        } catch (NumberFormatException e) {
            System.out.println("Error: Invalid account format.");
        } catch (SQLException e) {
            System.out.println("Database error: " + e.getMessage());
        }
    }
}

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        private static double getBalance(Connection conn, int accNum) throws
SQLException {
            String sql = "SELECT balance FROM accounts WHERE account_number = ?";
            try (PreparedStatement pstmt = conn.prepareStatement(sql)) {
                pstmt.setInt(1, accNum);
                try (ResultSet rs = pstmt.executeQuery()) {
                    if (rs.next()) {
                        return rs.getDouble("balance");
                    }
                }
            }
            return 0.0;
        }
    }
}
```